
Fractions, Decimals, and Percentages

Learning Outcome

When you complete this module you will be able to:

Perform basic arithmetic operations involving fractions, decimals and percentages.

Learning Objectives

Here is what you will be able to do when you complete each objective:

1. Identify proper and improper fractions and mixed numbers.
2. Add, subtract, multiply and divide fractions and reduce them to lowest terms.
3. Convert fractions to decimal numbers and decimal numbers to fractions.
4. Evaluate percentage problems.





INTRODUCTION

Fractions are represented by an integer above a line with another integer below the line. This is known as a common fraction. The top number is called the numerator and the bottom number is called the denominator.

Why are the words numerator and denominator used? Numerator originates from the Latin word numeratus, which means to count, and denominator is from the Latin word nominatus which means name. So a numerator indicates how many and the denominator indicates the name or size of units.

$$\frac{\text{numerator}}{\text{denominator}} = \frac{14}{37}$$

which in this case means that one complete unit would contain 37 equal pieces, but we are only considering 14 of them.

FRACTIONS

Fractions are numbers that can be interpreted in several ways. A fraction can be thought of as a ratio of two integers (i.e., $2/3$), as an indicated division (i.e., $3/4$ means $3 \div 4$), or as one or more parts of a unit (i.e., $5/11$ means a unit divided into 11 equal parts, of which 5 of the equal parts are considered).

Fractions can take a number of forms. Proper fractions are less than one; the numerator or integer on top, is less than the denominator, or integer on the bottom. Improper fractions are greater than one, with the numerator larger than the denominator. Mixed numbers are whole numbers with a fraction. Some typical examples of these three forms of fractions are:

proper fraction: $2/3$, $3/8$, $17/23$

improper fraction: $5/3$, $9/8$, $13/7$

mixed number: $2 \frac{1}{4}$, $9 \frac{7}{8}$, $15 \frac{3}{4}$

In the fraction $5/8$, 5 is the numerator and 8 is the denominator.

In the improper fraction $5/3$, 5 is the numerator and 3 is the denominator.

Converting Improper Fractions to Mixed Numbers

Divide the top by the bottom; the answer is a whole number and the remainder, if any, is placed over the same denominator, thus:

$$\frac{5}{2} = 2 \frac{1}{2}$$

$$\frac{19}{4} = 4 \frac{3}{4}$$

and

$$\frac{45}{4} = 11 \frac{1}{4}$$

Converting Mixed Numbers to Improper Fractions

Multiply the whole number by the denominator of the fraction, add on the numerator and place the result over the same denominator, thus:

$$2 \frac{2}{3} = \frac{(2 \times 3) + 2}{3}$$

$$= \frac{8}{3}$$

or

$$11 \frac{5}{8} = \frac{(11 \times 8) + 5}{8}$$

$$= \frac{93}{8}$$





Reduction of Fractions

A fraction has been reduced to its lowest terms when the numerator and denominator are prime to each other, that is when they have no common factors.

Example 1:

Reduce 75/105 to the lowest terms.

Solution:

$$\begin{aligned}\frac{75}{105} &= \frac{5 \times 15}{5 \times 21} \\ &= \frac{3 \times 5}{3 \times 7} \\ &= \frac{5}{7} \text{ (Ans.)}\end{aligned}$$

Equivalent Fractions

If both numerator and denominator of a fraction are multiplied by the same number, the value of the fraction remains unchanged. For example:

$$\begin{aligned}\frac{6}{7} &= \frac{6 \times 4}{7 \times 4} \text{ (multiplying top and bottom by 4)} \\ &= \frac{24}{28}\end{aligned}$$

If both numerator and denominator of a fraction are divided by the same number, the value of the fraction remains unchanged.

Thus

$$\begin{aligned}\frac{4}{12} &= \frac{2}{6} \text{ (dividing top and bottom by 2)} \\ &= \frac{1}{3} \text{ (dividing top and bottom by 2 again)}\end{aligned}$$

Example 2:

Reduce $9/12$ to fourths.

Solution:

$$12 \div 4 = 3$$

then:

$$\frac{9 \div 3}{12 \div 3} = \frac{3}{4} \text{ (Ans.)}$$

Addition of Fractions**1. Like Fractions**

When fractions have the same denominator, addition or subtraction is simple. The numerators are added or subtracted and the denominator remains the same; thus:

$$\begin{aligned} \frac{2}{5} + \frac{3}{5} &= \frac{5}{5} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \frac{1}{5} + \frac{1}{5} + \frac{3}{5} &= \frac{5}{5} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \frac{7}{9} - \frac{1}{9} &= \frac{6}{9} \\ &= \frac{2}{3} \end{aligned}$$





2. Unlike Fractions

Before adding or subtracting common fractions, they must be brought to the same denominator; thus:

$\frac{1}{8}$, $\frac{1}{4}$, and $\frac{1}{16}$ cannot be added directly.

They must be brought to the same denominator and then their numerators added, thus:

$$\frac{1}{8} = \frac{2}{16}, \quad \frac{1}{4} = \frac{4}{16}, \quad \frac{1}{16} = \frac{1}{16}$$

Adding the numerators: $2 + 4 + 1$ equals 7. The sum is therefore $\frac{7}{16}$.

Example 3:

$$\text{Add } \frac{4}{9} + \frac{7}{12} + \frac{13}{24}$$

Solution:

The LCM of 9, 12 and 24 is found by factoring each.

$$9 = 3 \times 3$$

$$12 = 2 \times 2 \times 3$$

$$24 = 2 \times 2 \times 2 \times 3$$

The LCM will be $2 \times 2 \times 2 \times 3 = 72$ and this is the lowest common denominator for these fractions.

Therefore:

$$\frac{4}{9} = \frac{32}{72}, \quad \frac{7}{12} = \frac{42}{72}, \quad \text{and} \quad \frac{13}{24} = \frac{39}{72}$$

and:

$$\begin{aligned} \frac{32 + 42 + 39}{72} &= \frac{113}{72} \\ &= 1 \frac{41}{72} \text{ (Ans.)} \end{aligned}$$

Subtraction of Fractions

The same principles apply in subtraction as in addition; the quantities are brought to the same denominator and then one of the numerators is subtracted from the other.

Example 4:

$$\begin{aligned}\frac{7}{8} - \frac{11}{64} &= \frac{56}{64} - \frac{11}{64} \\ &= \frac{45}{64} \text{ (Ans.)}\end{aligned}$$

Multiplication of Fractions

In multiplication of two or more fractions, all numerators must be multiplied together and placed above the product of all denominators. For example:

$$\begin{aligned}1\frac{1}{2} \times 2\frac{2}{3} \times \frac{7}{16} &= \frac{3}{2} \times \frac{8}{3} \times \frac{7}{16} \\ &= \frac{168}{96} \\ &= \frac{7}{4} \\ &= 1\frac{3}{4}\end{aligned}$$

Factors, however, which appear one in the numerator and one in the denominator may be cancelled out. In the last example:

$$\frac{3}{2} \times \frac{8}{3} \times \frac{7}{16}$$

the threes cancel out and 8 divides into 16 twice. The expression then becomes:

$$\begin{aligned}\frac{1}{2} \times \frac{1}{\cancel{3}} \times \frac{7}{\cancel{16}^2} &= \frac{7}{4} \\ &= 1\frac{3}{4}\end{aligned}$$



Division of Fractions

When dividing by a fraction, invert it and multiply.

Example 5:

$$\begin{aligned}\frac{3}{7} \div \frac{2}{5} &= \frac{3}{7} \times \frac{5}{2} \\ &= \frac{15}{14} \\ &= 1 \frac{1}{14} \text{ (Ans.)}\end{aligned}$$

This can also be written:

$$\begin{aligned}\frac{3/7}{2/5} &= \frac{3}{7} \times \frac{5}{2} \\ &= \frac{15}{14} \\ &= 1 \frac{1}{14} \text{ (Ans.)}\end{aligned}$$

Example 6:

$$\begin{aligned}\frac{2}{5} \div \frac{9}{11} &= \frac{2}{5} \times \frac{11}{9} \\ &= \frac{22}{45}\end{aligned}$$

Example 7:

$$\begin{aligned}2 \frac{1}{7} \div 1 \frac{1}{4} &= \frac{15}{7} \div \frac{5}{4} \\ &= \frac{15}{7} \times \frac{4}{5} \\ &= \frac{12}{7} \\ &= 1 \frac{5}{7} \text{ (Ans.)}\end{aligned}$$

Example 8:

$$\begin{aligned}
 2\frac{1}{2} \times 3\frac{2}{3} \div 1\frac{4}{7} &= \frac{5}{2} \times \frac{11}{3} \div \frac{11}{7} \\
 &= \frac{5}{2} \times \frac{11}{3} \times \frac{7}{11} \\
 &= \frac{35}{6} \\
 &= 5\frac{5}{6} \text{ (Ans.)}
 \end{aligned}$$

Example 9:

$$\begin{aligned}
 2\frac{1}{2} \div \frac{5}{16} &= \frac{\cancel{2}^1}{\cancel{2}_1} \times \frac{\cancel{16}^8}{\cancel{5}_1} \\
 &= 8 \text{ (Ans.)}
 \end{aligned}$$

Principles

The following principles may be stated for fractions:

1. Multiplying or dividing both numerator and denominator by the same number does not change the value of the fraction.
2. Multiplying the numerator or dividing the denominator by a number multiplies the fraction by that number.
3. Dividing the numerator or multiplying the denominator by a number divides the fraction by that number.



Decimal Numbers

A decimal number is a fraction that has 10 or a power of 10 as a denominator.

Thus $\frac{26}{100}$ or $\frac{76}{100}$ or $\frac{532}{10\,000}$ are decimal fractions.

In writing a decimal fraction, it is convenient to omit the denominator and indicate what it is by placing a point or period (.), called a decimal point, in the numerator so that there shall be as many figures to the right of this point as there are zeros in the denominator.

The above fractions now become 0.26, 0.76 and 0.0532, read as “decimal two-six, decimal seven-six, and decimal zero-five-three-two.

The term “common” is used to describe all fractions other than those which are decimal fractions.

It will be necessary from time to time to convert from one form of fraction to another. The techniques used are summarized, with an example given for each.

1. To change from a common fraction to a decimal fraction, simply divide the numerator by the denominator. On a calculator, this technique becomes quite straightforward.

$$\begin{aligned}\frac{4}{100} &= 4 \div 100 \\ &= 0.04\end{aligned}$$

2. To change a decimal fraction to a common fraction, write the nonzero portion of the decimal fraction as the numerator. Then write as the denominator, a one (1) with a zero for every digit, or place, after the decimal. Finally, simplify the numerator and denominator by cancellation.

$$\begin{aligned}0.125 &= \frac{125}{1000} \\ &= \frac{5}{40} \text{ (dividing numerator and denominator by 25)} \\ &= \frac{1}{8} \text{ (dividing numerator and denominator by 5)}\end{aligned}$$

PERCENTAGES

When quantities must be compared, fractions are not always helpful. Nor are decimals necessarily ideal for rapid calculations. In many cases, it is better to arrange matters so that all fractions considered have a denominator of 100.

The words per cent mean by the hundred and the symbol used to represent percent is %. Thus 10% means 10 per hundred and can be written as a common fraction, $10/100$ or as a decimal fraction, 0.10, representing one tenth of a whole number.

For example, 10% of 40 litres of oil would be:

$$\frac{10}{100} \times 40 = 4 \text{ litres}$$

Since a fraction is part of unity and a percentage is part of one hundred, a fraction can be converted into a percentage by multiplying by 100. Conversely, a percentage can be changed to a fraction by dividing by 100. Thus the common fraction $3/4$ becomes $(3 \times 100)/4 = 75\%$ or the decimal fraction 0.75 becomes $0.75 \times 100 = 75\%$ and vice versa, the percentage 75 becomes $75/100$ or 0.75.

To express percentage increase or decrease, the rules are:

$$\% \text{ increase} = \frac{\text{increase}}{\text{original amount}} \times 100$$

$$\% \text{ decrease} = \frac{\text{decrease}}{\text{original amount}} \times 100$$

Example 10:

A spring 12 cm long is stretched to 15 cm. Calculate the percent increase in length.

Solution:

$$\begin{aligned} \text{increase} &= \text{final length} - \text{original length} \\ &= 15 \text{ cm} - 12 \text{ cm} \\ &= 3 \text{ cm} \\ \% \text{ increase} &= \frac{\text{increase}}{\text{original length}} \times 100 \\ &= \frac{3}{12} \times 100 \\ &= 25\% \text{ (Ans.)} \end{aligned}$$





Percentage Problems

As an operator, you may have to deal with percentages in solving questions such as the following:

1. “This year’s gas consumption is what percentage of last year’s?”
2. “By what percentage did this year’s gas consumption increase over last year’s?”
3. “Last year’s gas consumption was lower than this year’s by what percent?”

The answers to the above questions are all different. For example, if this year’s gas consumption is $24 \times 10^6 \text{ m}^3$ and last year’s was $18 \times 10^6 \text{ m}^3$, then the answers to the above questions would be:

$$1. \quad \frac{24 \times 10^6 \text{ m}^3}{18 \times 10^6 \text{ m}^3} \times 100 = 133 \frac{1}{3} \%$$

$$2. \quad \frac{6 \times 10^6 \text{ m}^3}{18 \times 10^6 \text{ m}^3} \times 100 = 33 \frac{1}{3} \%$$

$$3. \quad \frac{6 \times 10^6 \text{ m}^3}{24 \times 10^6 \text{ m}^3} \times 100 = 25 \%$$

There are three basic types of percent problems:

1. To find a certain percentage of a number (n).
2. To find a number (n) when a percent of it is known.
3. To find what percent (p) one number is of another number.

Example 11:

12% of 500 is what number (n)?

Solution:

$$n = \frac{12}{100} \times 500$$

$$n = 12 \times 5$$

$$= \mathbf{60 \text{ (Ans.)}}$$

Example 12:

5% of what number is 10 ?

Solution:

$$\frac{5}{100} (n) = 10$$

$$\begin{aligned} n &= \frac{10 \times 100}{5} \\ &= \mathbf{200} \text{ (Ans.)} \end{aligned}$$

Example 13:

21 is what percent of 300 ?

Solution:

$$\frac{p}{100} \times 300 = 21$$

or

$$p = \frac{21}{300} \times 100$$

$$\begin{aligned} p &= \frac{21}{3} \\ &= \mathbf{7} \text{ (Ans.)} \end{aligned}$$





Self Test

After completion of the self-test, check your answers with the answer guide that follows the test.

1. Reduce the following fractions to lowest terms.
 - (a) $18/48$
 - (b) $21/39$
 - (c) $15/30$
 - (d) $25/64$
 - (e) $12/4$
2. Convert the following mixed numbers to improper fractions.
 - (a) $3 \frac{1}{4}$
 - (b) $4 \frac{7}{8}$
 - (c) $9 \frac{2}{7}$
 - (d) $1 \frac{1}{5}$
 - (e) $6 \frac{3}{4}$
3. Convert the following improper fractions to mixed numbers. Reduce the fraction portion to lowest terms.
 - (a) $9/4$
 - (b) $13/3$
 - (c) $18/8$
 - (d) $19/4$
 - (e) $11/5$
4. Add the following groups of fractions. Express your answer as a mixed number if required.
 - (a) $1/4, 3/16, 9/32$
 - (b) $11/13, 5/39, 2/3$
 - (c) $3/6, 1/8, 1/3$
 - (d) $9/5, 6/15, 26/75$
 - (e) $7/8, 11/120, 3/12$
5. Do the following subtractions. Convert your answer to mixed numbers in lowest terms if required.
 - (a) $15/4 - 1/3$
 - (b) $11/12 - 1/6$
 - (c) $13/15 - 3/5$
 - (d) $21/25 - 1/2$

-
- (e) $13/3 - 5/6$
6. Multiply the following fractions. Reduce your answer to lowest terms.
- (a) $3/4 \times 5/6 \times 2/3$
(b) $1/4 \times 1/2 \times 3/8$
(c) $9/16 \times 2/3$
(d) $3/5 \times 1/3 \times 3/4$
(e) $4/5 \times 1/5 \times 9/5$
7. Divide the following fractions. Reduce your answer to lowest terms.
- (a) $4/9 \div 5/6$
(b) $12/13 \div 3/4$
(c) $1/2 \div 6/7$
(d) $11/12 \div 1/4$
(e) $5/9 \div 2/3$
8. Convert the following fractions to decimals.
- (a) $3/5$
(b) $1/4$
(c) $17/100$
(d) $1/10$
(e) $9/16$
9. Convert the following decimals to fractions. Reduce your answer to lowest terms.
- (a) 0.11
(b) 0.80
(c) 0.375
(d) 0.45
(e) 0.76
10. Convert the following fractions or decimals to percent. Show only two decimal places.
- (a) $13/16$
(b) $29/45$
(c) $68/70$
(d) 0.192
(e) 0.356





Self Test Answers

- | | |
|---|--|
| 1. (a) $\frac{3}{8}$
(b) $\frac{7}{13}$
(c) $\frac{1}{2}$
(d) $\frac{25}{64}$
(e) 3 | 6. (a) $\frac{5}{12}$
(b) $\frac{3}{64}$
(c) $\frac{3}{8}$
(d) $\frac{3}{20}$
(e) $\frac{36}{125}$ |
| 2. (a) $\frac{13}{4}$
(b) $\frac{39}{8}$
(c) $\frac{65}{7}$
(d) $\frac{6}{5}$
(e) $\frac{27}{4}$ | 7. (a) $\frac{8}{15}$
(b) $1\frac{3}{13}$
(c) $\frac{7}{12}$
(d) $3\frac{2}{3}$
(e) $\frac{5}{6}$ |
| 3. (a) $2\frac{1}{4}$
(b) $4\frac{1}{3}$
(c) $2\frac{1}{4}$
(d) $4\frac{3}{4}$
(e) $2\frac{1}{5}$ | 8. (a) 0.60
(b) 0.25
(c) 0.17
(d) 0.10
(e) 0.5625 |
| 4. (a) $\frac{23}{32}$
(b) $1\frac{25}{39}$
(c) $\frac{23}{24}$
(d) $2\frac{41}{75}$
(e) $1\frac{13}{60}$ | 9. (a) $\frac{11}{100}$
(b) $\frac{4}{5}$
(c) $\frac{3}{8}$
(d) $\frac{9}{20}$
(e) $\frac{19}{25}$ |
| 5. (a) $3\frac{5}{12}$
(b) $\frac{3}{4}$
(c) $\frac{4}{15}$
(d) $\frac{17}{50}$
(e) $3\frac{1}{2}$ | 10. (a) 81.25%
(b) 64.44%
(c) 97.14%
(d) 19.20%
(e) 35.60% |